Experiment 10.1

Analysis of Critical Frequency and Maximum Usable Frequency (MUF) using MATLAB

Aim

To analyse the critical frequency of the ionosphere and the maximum usable frequency (MUF) for different angles of incidence.

Software Required

MATLAB (MATLAB Online)

Procedure / Algorithm

- Take electron density Nmax = 0.1 to 1.0 ($\times 10^{12}$ electrons/m^3) and compute fc = $9\sqrt{Nmax}$.
- Take fixed fc = 6.3 MHz, angle of incidence 0° to 70°, and compute MUF = $fc \cdot \sec(\theta)$.
- Compute the MUF/fc ratio and plot all the results.

Output

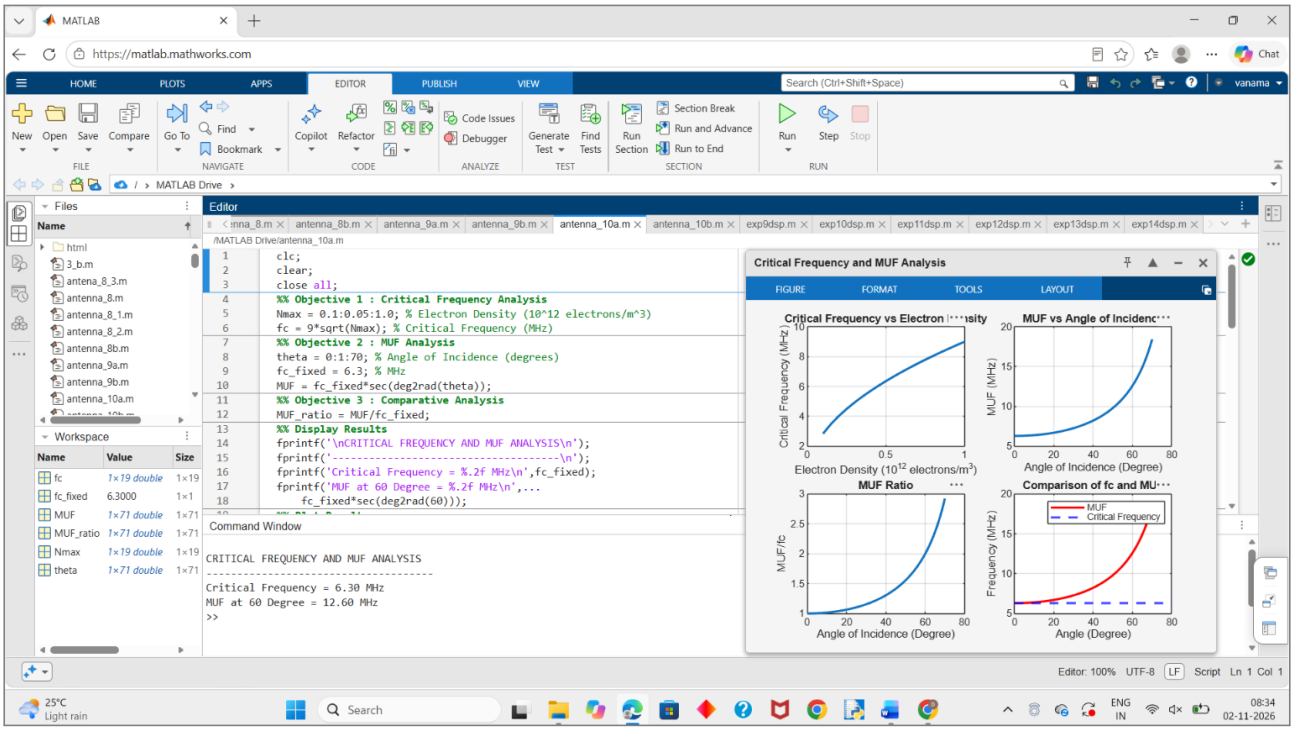


Fig. 10.1 Critical frequency and MUF analysis

Result

- Critical frequency = 6.30 MHz.
- MUF at 60° = 12.60 MHz.
- MUF rises with the angle of incidence and critical frequency rises with electron density.